Engineering
ENGINEER 2PX3
Communication and Social Impact
Winter 2026

INSTRUCTOR INFORMATION

Mr. Kyle Ansilio (Course Coordinator, Lecture Professor)

Dr. Zeinab Hosseinidoust (Lecture Professor)

Dr. Sam Scott (Lecture Professor)

Dr. Shelir Ebrahimi (Design Studio Professor)

Dr. Duncan Cree (Design Studio Professor)

Dr. Joseph Kish (Design Studio Professor)

Dr. John Preston (Design Studio Professor)

Dr. Matiar Howlader (Design Studio Professor)

All instructors may be reached at prof2px3@mcmaster.ca. Office hours are starting in Week 2 (January 12-16, 2025).

Deepam: Wednesdays, 3-4 p.m., ETB 237

Refah: Thursdays, 2 - 3 p.m., ETB 237

ADDITIONAL STAFF

Mr. Basem Yassa Instructional Support Team Lead yassab@mcmaster.ca	**Mr. Brendan Fallon** Instructional Assistant II fallonbr@mcmaster.ca	**Ms. Refah Nanziba** Instructional Assistant Intern nanzibar@mcmaster.ca	**Mr. Deepam Patel** Instructional Assistant Intern pated223@mcmaster.ca

COURSE INFORMATION

Lectures:
C01: Monday and Wednesday at 11:30 am – 12:20 pm PGCLL 127
C02: Monday and Wednesday at 8:30 am – 9:20 am LRW B1007
C03: Tuesday and Thursday at 11:30 am – 12:20 pm CNH 104

Design Studio:
T01: Friday at 2:30 pm – 4:20 pm
T02: Thursday at 8:30 am – 10:20 am
T03: Tuesday at 1:30 pm – 3:20 pm
T04: Thursday at 3:30 pm – 5:20 pm
T05: Wednesday at 12:30 pm – 2:20 pm
T06: Tuesday at 3:30 pm – 5:20 pm

T08: Thursday at 11:30 am – 1:20 pm

All tutorials take place in ETB 235/236 and ETB 238/239.

Course Dates: 01/05/2026 - 04/22/2026
Units: 3.0
Course Delivery Mode: In-Person. In-person attendance is required for this course.
Course Description: A multidisciplinary integrated learning course focused on communication and the engineering design cycle. Formal and informal, verbal and written communication skills are fostered through project-based design projects.

IMPORTANT LINKS

- Mosaic
- Avenue to Learn
- Student Accessibility Services - Accommodations
- McMaster University Library

Top Hat Lecture Sections:

- C01
- C02
- C03

COURSE LEARNING OUTCOMES

1. Perform different steps of the engineering design cycle.
2. Effectively communicate the design process and decisions.
3. Apply principles to demonstrate respectful and productive teamwork.
4. Articulate insights into the design process and team dynamics.

COURSE LEARNING GOALS

Effective communication skills are essential for success in the engineering profession. In a field driven by innovation and collaboration, engineers must convey complex ideas and technical information with clarity and precision to a range of audiences. Written communication skills enable them to articulate their thoughts in formal and informal documentation, ensuring seamless knowledge transfer and efficient project management. Similarly, strong oral communication skills empower engineers to present, collaborate, persuade, and inform a broad audience (e.g., stakeholders, colleagues, clients, supervisors, etc.). Clear and concise communication fosters understanding, promotes teamwork, and builds trust, ultimately enabling engineers to deliver innovative solutions that drive progress in our interconnected world. This course aims to develop technical communication skills through a series of multidisciplinary design projects.

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national quality standard and cover essential educational requirements. Graduate Attributes are qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

4.1 Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions. (Learning Outcome 1)
4.4 Justifies and reflects on design decisions, giving consideration to limitations, assumptions, constraints and other relevant factors. (Learning Outcome 1)
6.2 Manages interpersonal relationships, taking leadership responsibilities as needed. (Learning Outcome 3)
7.2 Composes an effective written document for the intended audience. (Learning Outcome 2)
7.3 Composes and delivers an effective oral presentation for the intended audience. (Learning Outcome 2)
8.1 Describes the duty of a Professional Engineer to the public, client, employer, and the profession. (Learning Outcome 4)
9.1 Evaluates the environmental impact of engineering activities, identifies uncertainties in decisions, and promotes sustainable design. (Learning Outcome 2)
9.2 Evaluates the social impact of engineering activities, including health, safety, legal, cultural, and other relevant factors, and identifies uncertainties in decisions. (Learning Outcome 2)
10.2 Applies the principles of equity and universal design to ensure equitable treatment of all stakeholders. (Learning Outcomes 2, 4)
12.1 Critically assesses one's own educational needs and opportunities for growth. (Learning Outcome 4)

Course Schedule

Week	Lecture 1	Lecture 2
1	Course Introduction	Engineering Design
2	Evaluation Framework	Evaluation Framework (cont.)
3	Written Communication I	Written Communication II
4	Oral Communication I	Problem Statements
5	Brainstorming	Stakeholder Analysis
6	Reflection on Equitable Design	Team Dynamics & Roles + Personality Types
	Reading Week	
7	Oral Communication II	Oral Communication III
8	Project Management	Critical Path Method
9	Written Communication III	Written Communication IV
10	Inclusive Design	Data Visualization
11	Presentations	Presentations
12	Presentations	Presentations
13	Presentations	Presentations

Note: The above structure represents a plan and is subject to adjustment term by term. The instructor and the university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

Project	**Design Steps**	**Duration**
Project 1	Evaluate	3 weeks
Project 2	Ask & Define + Conceptualize + Evaluate + Improve	9 weeks

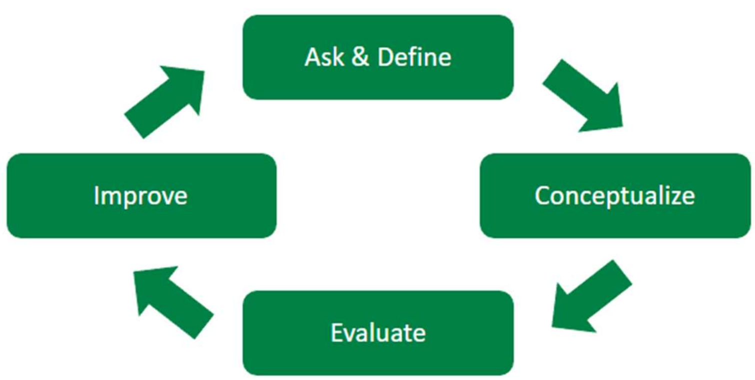
REQUIRED MATERIALS AND TEXTS

See Course Schedule for Assigned Readings available on Avenue.

CLASS FORMAT

The course is scheduled for two in-person one-hour lectures per week, along with one in-person two-hour Design Studio per week. Classes will consist of an interactive lecture format by the instructor, with students having the opportunity to ask questions concerning the information being provided to them. The lectures will be recorded, and students are strongly encouraged to attend the in-person lectures. In addition to formal class time, students are invited to meet with the instructor by appointment to discuss course content or any other concerns.

Engineering design looks very different from one engineering discipline to another. For this course, the focus will be on the high-level design cycle below:



This course is delivered in two design projects. Each project in 2PX3 focuses on different steps of the design cycle. Project 1 is an individual project, while Project 2 is a group project. Each project includes a report that is submitted as the project concludes. Project 2 includes a peer evaluation that assesses students' contributions to the project.
All students are expected to use MS Teams or Outlook as their primary communication tool when working on project activities. MS Teams and Outlook conversations are tied to your student MacID, thus ensuring your identity when managing and resolving conflict that may arise in a team-based design project. Should team conflicts arise, other forms of social media will not be reviewed under any circumstances.

COURSE EVALUATION

Component	Weight (%)	Grade Type	Due Date/Occurrence
Project 1 Report	15%	Individual	Feb. 2nd, 2026
Reflections (2)*	10%	Individual	Feb. 11th & Mar 16th, 2026
Mock Interview *	5%	Individual	Feb 2nd – Feb 27th, 2026
3 Minute Pitch *	5%	Individual	Mar. 2nd – Mar. 20th, 2026
Project 2 Video	10%	Group	Mar. 2nd , 2026
Project 2 Final Presentation **	15%	Group	Mar. 30th – Apr. 10th, 2026
Project 2 Report **	40%	Group	Mar. 30th, 2026
Total	100%		

* 5% each, subject to attendance regarding scheme described below.
**Group marks subject to individual penalties.

COURSE EVALUATION DETAILS

<u>Attendance Reweight</u>

Attendance will be taken during both lecture and Design Studio using the Top Hat app. Students should sign up for Top Hat using their McMaster emails to ensure that they are able to access course materials. For students that maintain an attendance rate of 75% or higher for both lectures and Design Studios, the weights of the assessments will be shifted to benefit students in the following ways:

- Half of the weight of the lowest Reflection grade (5.0%) will be shifted to the weight of the highest Reflection grade.
- Half the weight of the lowest Interview/3 Minute Pitch grade (5.0%) will be shifted to the highest.

Students may review their attendance rate at any time by accessing the *Gradebook* section in the Top Hat application. This section provides a record of all attendance submissions along with the total number of attendance requests issued during the course.

Please note: any individually graded assessment that is dropped as a result of a McMaster Student Absence Form (MSAF) will count as the dropped assessment for the Attendance Reweight. For example, if a student meets the Attendance Reweight and uses an MSAF on one of the Reflections, the other Reflection will count toward the student's final grade.

GRADING CONCERNS

Grades will be posted to Avenue as soon as possible upon completion.

- Any questions/concerns must be addressed electronically through an MS Form: ENGINEER 2PX3 Marking Concerns Form
- Concerns with regards to grading will not be considered without submitting through the appropriate channels (i.e., the posted MS Form).

GRADING SCALE

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

CONDUCT EXPECTATIONS

As a McMaster graduate student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

Students must be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in university activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

TURNITIN.COM

Some courses may use a web-based service (Turnitin.com) to reveal the authenticity and ownership of student-submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., online search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

GENERATIVE AI: SOME USE PERMITTED

Students may use generative AI in this course so long as the use of generative AI is referenced and cited following citation instructions given in the course. Students are not permitted to use generative AI to produce original work for any assessment. All prompts and outputs must be included in the appendix of the submission, along with a description of how the tool was used and must be cited appropriately. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the expectations for citation and reference and to do so appropriately. Please refer to **Guidelines on the Use of Generative AI in Teaching and Learning** for further information on this policy.

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. The academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.** Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. a grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/.
The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

THREE STRIKE SYSTEM

The three-strike system is a method used to enforce that all members contribute equally to the Design Projects and on the chance that a team member is refusing to cooperate or complete their task, then the strike system will be enforced. Strikes and their consequences are detailed below:

1. **First Strike:** A warning from the TA assigned to your group.
2. **Second Strike:** A second warning issued by your Design Studio IAI.
3. **Third Strike:** Leads to a consultation with a professor, who will assess your involvement in the project.
    - Possible outcomes include:
        - A penalty of up to **100% of the project weight**, or
        - An assignment of **replacement work**.

If you have three noted absences in a project, this leads to an automatic three strikes and a consultation with a professor. In the event that there are underlying circumstances that may prevent a team member from completing their work, please communicate said circumstances with your team and with your Design Studio IAI to prevent you from acquiring a strike. If you are having issues with team members, it is your responsibility to communicate early and often with your TA when they come around in Design Studio.

Examples of strikes (this IS NOT an exhaustive list):

1. Did not fulfill individual responsibilities outside of Design Studio.
2. Being present in Design Studio but not contributing.
3. Being absent in Design Studio and not informing team members.

COURSES WITH AN ON-LINE ELEMENT

In this course, we will be using Avenue to Learn. Students should be aware that, when they access the electronic components of this course, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation of this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

Students must be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in university activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

EQUITY, DIVERSITY & INCLUSION

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, particularly if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or the Equity and Inclusion Office.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s Academic Accommodation of Students with Disabilities policy.

It is the prerogative of the instructor to determine the appropriate accommodation for term work in this course based on the student’s approved SAS plan.

- For written and recorded assessments, deadlines will be extended by up to **1 week.** This includes the **Project 1 Report** and both **Reflections**.
- Individual In-person assessments requiring accommodation will be rescheduled to **Week 11** of the course (Mar. 23rd – Mar. 27th, 2026). This includes the **Mock Interview** and the **3-Minute Pitch**.
- Team deliverables **CAN NOT** be accommodated. This includes the **Project 2 Video**, **Project 2 Report**, and the **Project 2 Final Presentation**.

A summary of these accommodations may be found below under the Summary of SAS Accommodations and MSAF Actions section below. It's the student's responsibility to activate their accommodations at their earliest convenience and before due dates for assignments. Accommodations cannot be made retroactively for due dates that have passed prior to the activation of an accommodation plan.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

- All MSAFs are to be directed to engic@mcmaster.ca, not to a professor in this course. Sending to another email address will delay processing.
- If you do not contact engic@mcmaster.ca after submitting your MSAF, your MSAF may not be processed.
- It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in his/her course.
    - For written and recorded assessments, the assessment deadline will be extended by **1 week**. This includes the **Project 1 Report** and both **Reflections**.
    - MSAFs for the **Mock Interview** and **3 Minute Pitch** will be rescheduled to **Week 11** of the course (Mar. 23rd – Mar. 27th, 2026).
    - Team deliverables **CAN NOT** have an MSAF applied to them. This includes the **Project 2 Video**, **Project 2 Report**, and **Project 2 Final Presentation**.

A summary of these accommodations may be found in the table below.

SUMMARY OF SAS ACCOMMODATIONS AND MSAF ACTIONS

Assignment	Due Date/Occurrence	SAS Accommodation	MSAF Action	Latest Acceptable Submission
Project 1 Report	Feb. 02nd, 2026	1 Week Extension	1 Week Extension	Feb. 09th, 2026
Reflection 1	Feb. 11th, 2026	1 Week Extension	1 Week Extension	Feb. 18th, 2026
Project 2 Video	Mar. 02nd, 2026	Unapplicable	Unapplicable	
Reflection 2	Mar. 16th, 2026	1 Week Extension	1 Week Extension	Mar. 23rd, 2026
Project 2 Report	Mar. 30th, 2026	Unapplicable	Unapplicable	
Mock Interview	Feb 2nd – Feb 27th, 2026	Reschedule to Week 11	Reschedule to Week 11	Mar. 23rd – Mar. 27th, 2026
3 Minute Pitch	Mar. 2nd – Mar. 20th, 2026	Reschedule to Week 11	Reschedule to Week 11	Mar. 23rd – Mar. 27th, 2026
Project 2 Final Presentation	Mar. 30th – Apr. 10th, 2026	Unapplicable	Unapplicable	

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS, OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should normally submit their request to their Faculty Office within 10 working days of the beginning of the term in which they anticipate a need for accommodation or to the Registrar's Office before their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright-protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by university instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor or any authorized personnel for authorized distribution, or by a student for personal study. Students should be aware that their voices and/or images may be recorded by others during the class, tutorial, or Design Studio. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.